Conditional statement: Sometimes we need to execute a statement only if a condition is true and do nothing if the condition is false. This is equivalent to the alternative statement in which the else-clause is empty. This variant of alternative statement is called a conditional statement. If C is a condition and S is a statement, then

```
if  C
   S
```

is a statement, called a conditional statement, that describes the following action:

1. Test whether C is true or false.
2. If C is true then do S; otherwise do nothing.

The conditional control flow is depicted in the flowchart of Figure 2.5.

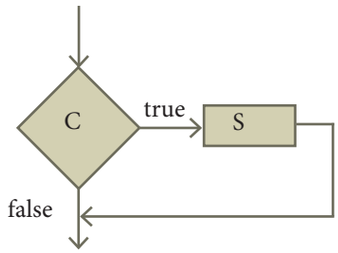


Figure 7.5: Conditional control flow

Example 7.2. Minimum of two numbers: Given two numbers a and b, we want to find the minimum of the two using the alternative statement. Let us store the minimum in a variable named result. Let a ↓ b denote the minimum of a and b (for instance, 4 ↓ 2 = 2, —5 ↓ 6 = -5). Then, the specification of algorithm minimum is

```
minimum(a, b)
-- input s : a , b
-- outputs: result = a ↓ b
```

Algorithm minimum can be defined as

```
1.   minimum(a, b)
2.      -- a, b
3.      if a < b
4.         result : = a
5.      else
6.         result = b
7.      --  result = a ↓ b
```

7.2.3 Case analysis

Alternative statement analyses the problem into two cases. Case analysis statement generalizes it to multiple cases. Case analysis splits the problem into an exhaustive set of disjoint cases. For each case, the problem is solved independently. If C1, C2, and C3 are conditions, and S1, S2, S3 and S4 are statements, a 4-case analysis statement has the form,

```
1. case  C1
2.    S1
3. case  C2
4.    S2
5. case  C3
6.    S3
7. else
8.    S4
```

The conditions C1, C2, and C3 are evaluated in turn. For the first condition that evaluates to true, the corresponding statement is executed, and the case analysis statement ends. If none of the conditions evaluates to true, then the default case S4 is executed.

1. The cases are exhaustive: at least one of the cases is true. If all conditions are false, the default case is true.
2. The cases are disjoint: only one of the cases is true. Though it is possible for

Chapter 7 Page 088-101.indd 92 3/24/2020 9:14:50 AM